Privacy Impact Assessment

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Executive Summary

System Description

This Privacy Impact Assessment (PIA) examines the deployment of Flock Safety automated license plate recognition (ALPR) cameras in a residential community setting. These cameras capture images of passing vehicles and extract license plate numbers along with related information such as vehicle characteristics, time stamps, and location data. The system is commonly used by homeowners' associations (HOAs) and law enforcement agencies to help investigate crimes and identify vehicles associated with suspicious activity.

Key Privacy Risks

Because the system continuously collects vehicle data, it creates the possibility of tracking individuals' movements over time. When license plate records are aggregated, they can reveal patterns such as residents' daily routines, when individuals leave or return home, and frequent visits by certain vehicles to specific houses. Over time, this data could also indicate social associations and identify regular visitors such as friends, service providers, or delivery drivers. There are also concerns about who can access this data and how it may be used. If data is widely shared across agencies or accessed without proper oversight, it could lead to misuse or surveillance beyond the system's original purpose.

Mitigations and Safeguards

Several safeguards can help reduce these risks. These include limiting how long data is stored, implementing role-based access controls so only authorized users can access the system, and maintaining audit logs to track how data is queried. Transparency is also important. Informing residents about camera locations and explaining how the data is used can help build trust. Additional oversight and clear limits on inter-agency data sharing would further reduce the potential for misuse.

Overall Privacy Risk Assessment

Overall Privacy Risk Level: High

While the system may provide benefits for public safety and crime investigations, the large-scale collection of location-linked vehicle data raises significant privacy concerns. In particular, the ability to track movement patterns over time creates risks related to surveillance, data aggregation, and most particularly, dark patterns.

Recommendations

If the system is deployed, it should be accompanied by clear governance policies and strong oversight. This includes transparent communication with residents, strict limits on how long data is retained, and clear policies regulating law enforcement access. Implementing these safeguards is important to ensure that public safety benefits are balanced with the protection of individual privacy.

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Introduction and Administrative Information

Author Information

This analysis is a course project for the Privacy Technology, Policy, and Law Class taught by Professor Ana Anton at the Georgia Institute of Technology. The four authors, Tom Kim, Shreya Mahesh, Naya Patel, and Aesha Shah[1], are equal contributors. The PIA is conducted by this four person team with defined roles including a lead responsible for overall structure and synthesis, a technical analyst that documents system functionality and data flows, a policy and governance analyst responsible for evaluating access controls and retention, and a stakeholder and risk analyst responsible for identifying impacted groups and assessing ethical risks.

Brief Description

The Privacy Impact Assessment evaluates Flock Safety (hereafter abbreviated to Flock) Automated License Plate Recognition (ALPR) and video analytics systems deployed in residential communities governed by individual residential community management, Homeowners Association (HOA) or other (hereafter shortened to HOA[2]). The system is provided by Flock and consists of cameras connected through network infrastructure and cloud-based software that captures, analyzes, and stores the license plate data [1].

The intended purpose of the Flock system is to ensure community safety and support incident investigations. The system is designed to deter property related crime, provide investigative leads following reported incidents, and to allow authorized personnel to search for vehicle detections associated with specific license plates. The HOA and residents are the primary beneficiaries of the deployment as they have sought out and implemented the Flock system for improved safety for residents and security within the community [1].

Scope of the Privacy Impact Assessment

The scope of this PIA includes the collection of vehicle images and license plate data at community entry and exit points or other designated locations, the generation of associated metadata such as timestamps and camera location, the transmission and storage of data in a vendor managed cloud environment, administrative access and search capabilities through a web based dashboard, and any sharing of data outside the originating organization. The assessment focuses on privacy impacts to residents, visitors, and service providers whose vehicle and location data may be captured when entering or traveling through the community. HOA and/or other management data systems that are not integrated with the Flock deployment, privately owned surveillance devices, and comprehensive jurisdiction legal analysis beyond high-level U.S. considerations are out of scope for this PIA.

[1] Alphabetically ordered

[2] Not all residential community managements are referred to as homeowners associations (HOAs), however, we are using the abbreviation HOA for simplicity.

System Description and Deployment Context

Overview of the Residential Deployment

In this residential deployment scenario, the Flock ALPR system is installed and operated by the HOA to monitor vehicle traffic entering and exiting the community. Flock designs and deploys fixed ALPR systems that capture license plate images and query associated vehicle data for safety and investigations [2]. Although this system is operated within a private residential setting, cameras are typically placed at entry and exit points and therefore may intersect with public roads, capturing information belonging to residents, guests, delivery drivers, and other service providers. As a result, in order to flag anomalies, the system extensively records ordinary daily vehicle movements.

System Functionality

The system operates through continuous, automated detection of vehicles as they drive within a camera's field of view. Each detection event generally includes a still image of the vehicle[3] , a focused image of the license plate, extracted license plate text using optical character recognition (OCR), and associated timestamp and location metadata [2]. Depending on configuration, additional vehicle details such as make, model, and color may also be generated.

The system is not designed to perform facial recognition and does not intentionally capture biometric identifiers such as race or gender. Instead, it focuses on vehicle attributes that are observable in public spaces [3]. Detection records are encrypted and transmitted to a vendor managed cloud infrastructure, where they are indexed and stored for later retrieval.

Authorized users access a web-based dashboard that enables search by license plate number, date and time range, and camera location. The system supports role-based access control and maintains logs of user queries to facilitate auditing and authorized use [4]. The design allows detection records to be exported or shared with authorized entities, including contracted security personnel or law enforcement, consistent with organizational policy.

Technical Architecture

At a high level, the system consists of fixed position cameras, a secure transmission layer, cloud storage and indexing infrastructure, and an administrative interface. Cameras capture vehicle images and perform limited preprocessing before securely transmitting encrypted detection events to Amazon Web Services (AWS) servers [1]. The cloud platform stores records in a centralized database and indexes them to enable efficient search queries.

[3] Based on our analysis, this image may also incidentally capture the driver, passengers, alterations to the vehicle / decals, vehicle make, vehicle model, vehicle color, etc.

There are trust boundaries that exist between the HOA (data controller[4]) and Flock's (data processor) use of the AWS cloud environment, between the HOA and any contracted private security providers that have been granted access, and between the HOA and external entities such as law enforcement if the data sharing is mandated by law or provided voluntarily[5] by the HOA. These boundaries are relevant for evaluating data governance and access control risks.

Physical Deployment Locations

In a residential context, cameras are typically installed at primary entrances and exits to maximize coverage of inbound and outbound vehicle traffic with secondary access points such as service entrances also monitored on occasion. Because these entrances often connect directly to public roads, the cameras may incidentally capture non-resident vehicles as well (for example, right-most or turning lanes).

The placement of cameras at perimeter checkpoints ensures collection of longitudinal movement records that reflect patterns of entry and exit. Over time these records could be used to retrospectively reconstruct travel histories, which contributes both to investigative usefulness and to privacy concerns.

Operational Workflow

The system functions through consistent passive data collection. Cameras record vehicle detections throughout the day and night and transmit that data to the cloud platform without requiring human monitoring. In routine operation, these stored records are dormant (unused) unless queried.

When a safety related incident occurs such as theft, vandalism, or suspicious activity, an authorized administrator identifies the relevant timeframe and conducts a targeted search using the recorded data. Detection events that match the criteria are reviewed and shared with law enforcement if appropriate under HOA policy.

Administrative responsibilities include managing user access credentials, configuring retention settings, and reviewing query logs where supported. These governance practices significantly influence the system's privacy risk profile.

Stakeholders and Institutional Context

Directly affected stakeholders include residents whose vehicles are (repeatedly) recorded, household members and relatives, other frequent guests, delivery drivers, and contractors. Because the system creates timestamped vehicle detection records, it enables retrospective analysis of vehicle movement within the community.

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[4] It has come to our attention while writing this analysis that the HOAs are not the only data controllers, to some extent, Flock also acts as a data controller.

[5] We assume HOA regulations governing voluntary information sharing with law enforcement are outlined in community covenants and contracts, with clear guidelines that protect residents and prevent disclosure without explicit approval from those involved.

Indirectly affected stakeholders include pedestrians and drivers near monitored entrances and individuals on adjacent public roadways whose vehicles may be incidentally scanned. With these cameras, the HOA's goals are to deter crime, improve investigative efficiency, and enhance residents' perception of safety. However, those benefits come with the accumulation of structured, searchable vehicle movement data, which forms a foundation for privacy risks that are evaluated in subsequent sections of this PIA.

Data Collection and Use

Personal Information

The Flock Safety system automatically collects license plate numbers, vehicle make, model, and color through still photographic images captured by fixed cameras placed at neighborhood entry and exit points. Each detection is associated with a time and date stamp as well as geolocation coordinates tied to the camera's position, allowing the system to record when and where a vehicle was observed [5]. This information has been used in investigations to help identify stolen vehicles, track suspect vehicles, and assist in urgent situations such as Amber Alerts. Although the system is designed to focus primarily on vehicles rather than individuals, there remains the possibility of incidental capture of people who are near the vehicle at the time the image is taken.

In addition to vehicle identifiers, the system generates metadata such as timestamps and location data that provide context for each detection. While these tags may appear relatively harmless on their own, repeated scans over time can reveal patterns about an individual's movements and routines. When aggregated and analyzed together, these data points may reveal more information about an individual's daily activities than one might think [6].

Table 1 illustrates how individual ALPR data elements (including but not limited to license plate numbers, timestamps, and geolocation) may appear non-sensitive in isolation but become privacy-invasive when aggregated over time to generate movement patterns and association insights.

Data Type	How It Is Collected	What It Reveals	Stored?	Who Can Access	Potential Privacy Impact
License Plate Number	Automatically captured via fixed camera image	Vehicle identity	Yes (cloud database)	HOA admins, law enforcement	Identifies vehicle owner when linked to DMV records
Vehicle Make/Model/Color	Image-based recognition	Vehicle characteristics	Yes	HOA admins, law enforcement	Supports vehicle identification and tracking
Still Photographic Image	Camera capture	Vehicle + possible individuals	Yes	Authorized users	May capture faces, bystanders, surroundings
Timestamp (Date/Time)	System-generated metadata	Movement timing patterns	Yes	Authorized users	Enables routine behavior analysis

Geolocation (Camera Location)	System metadata	Location of vehicle detection	Yes	Authorized users	Enables mapping of travel history
Movement Patterns	Derived from repeated detections	Travel routines, work/home location	Yes (derived analytics)	Law enforcement, possibly cross-agency	Reveals daily habits and private life
Association Data	Derived from vehicles appearing together	Social relationships	Derived	Law enforcement	Suggests networks of association
Visit Patterns	Derived from repeated presence at specific sites	Religious, medical, political activity	Derived	Law enforcement	Reveals sensitive affiliations

Sensitive Information

Although the Flock Safety system does not directly collect traditionally sensitive data, it can still be inferred through the aggregation of vehicle detection data. Because the system continuously scans and records license plate detections over time, it can generate detailed location histories that reveal routine travel patterns and behavioral trends. These patterns may allow analysts to reconstruct where a vehicle frequently travels and when those visits occur [7].

From these patterns, it may be possible to infer highly sensitive information about individuals. For example, repeated instances of the same license plate recorded when entering or exiting a residential community at consistent times may suggest affiliation to residents or indicate a resident's predictable routine. Additionally, vehicles that consistently appear together at the same locations or times may indicate social relationships or association networks. While these insights are not directly collected by the system, they can emerge through analysis of the accumulated data over a longer period of time [6].

Information Incidentally Captured

Because the Flock Safety system passively scans and records license plates of vehicles passing by, entering, or leaving monitored areas, it collects data about many individuals who are not suspected of any wrongdoing. Unlike systems where users voluntarily provide information, vehicle data is captured automatically whenever a car passes within range of the camera [8].

As a result, the system may incidentally collect information about a wide range of individuals, for example, students traveling to school, patients visiting medical facilities, journalists reporting on events, protesters attending demonstrations, undocumented individuals, neighborhood residents, delivery drivers, and general bystanders. These individuals may have no direct connection to a criminal investigation but are still included in the system's dataset simply because their vehicles passed a monitored location.

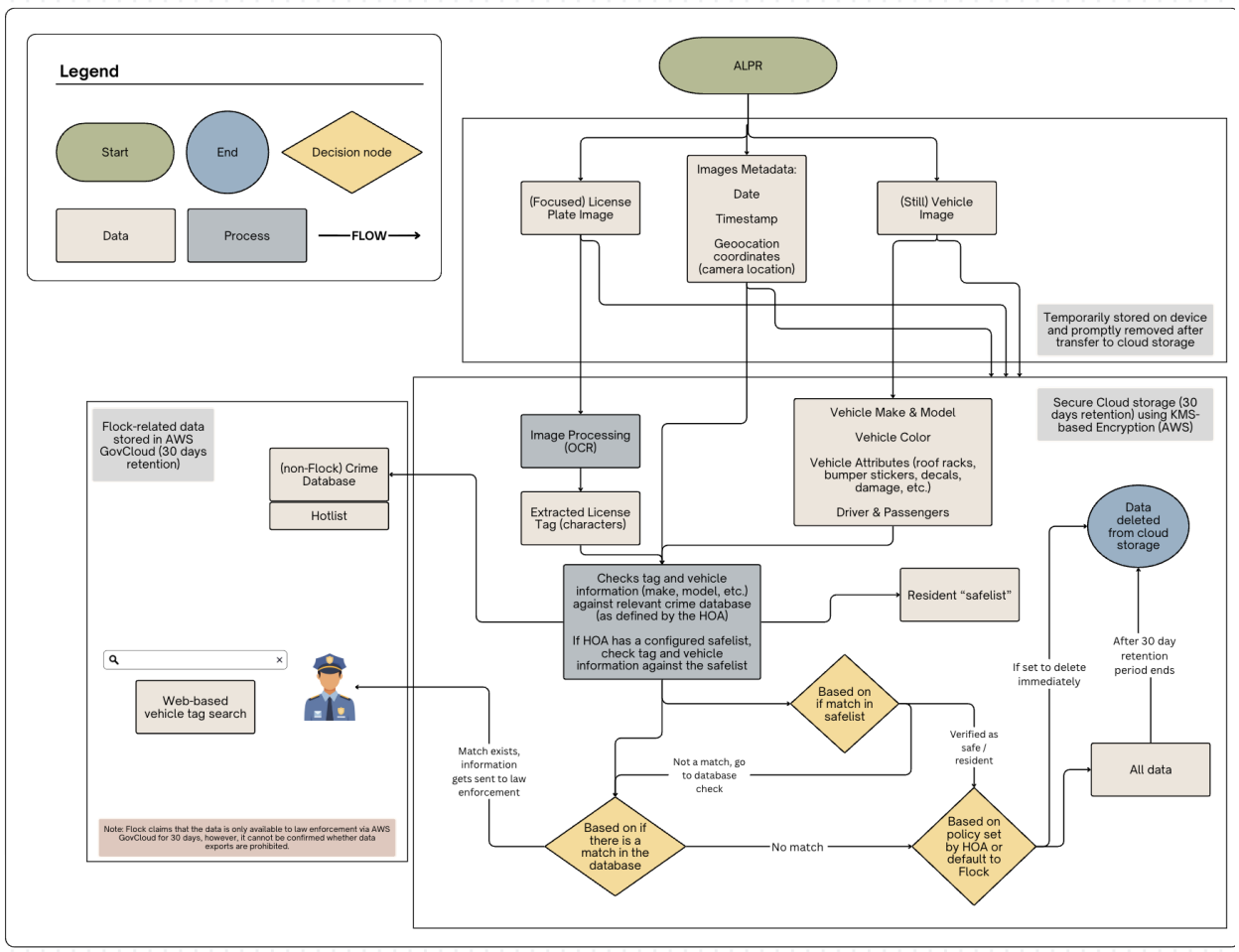
Since the data collection occurs automatically, individuals whose vehicles are recorded typically do not provide consent and may not even be aware that their movements are being captured. Consequently, the scope of surveillance extends beyond its intended investigative targets and affects a much broader population than those suspected of criminal activity.

Data Collection Benefits and Beneficiaries

The primary purpose of the Flock Safety system is to enhance public safety, deter crime, and improve investigative efficiency within neighborhoods and homeowner associations (HOAs). Based on user testimonials from Flock's website, residential communities have had good experiences with the system. In Oceanside, California, a neighborhood called Pacific Ridge was being targeted by break-ins. Omar Herrera, Head of Safety Committee at Pacific Ridge HOA said "Our whole community was being hit a lot. It was an ongoing issue with neighbors who were getting their houses broken into. We were being targeted by a new gang that was terrorizing the area." Local law enforcement recommended implementing Flock in the Pacific Ridge community so they did. They deployed a multi-layered, intelligence-driven security strategy to deter crime and strengthen their relationship with law enforcement. As a result, the Pacific Ridge community witnessed dramatic security improvements within months. According to their website, zero break-ins were reported in the 12 months following installation, down from 4 break-ins annually pre-Flock. Stolen vehicles previously spotted entering the area have disappeared completely, thanks to deterrence and criminal casing was stopped as the presence of Flock cameras discouraged scouting via public park trails. [9]

Data Flow Diagram

Figure 1 illustrates the data flow within the Flock ALPR deployment in a residential HOA context.



References: [13][14][15]

Walkthrough

Data collection begins when a vehicle enters the camera's field of view. The system captures:

- A focused image of the license plate
- A still image of the vehicle
- Metadata including timestamp and geolocation

The license plate image undergoes on-device or cloud-based OCR processing, generating extracted license tag characters. Vehicle attributes such as make, model, color, and distinguishing features may also be derived from the image [13].

All detection data is transmitted to secure cloud storage hosted on AWS, where it is encrypted and retained for a default period of 30 days unless modified by policy [13]. Within the cloud environment, extracted license tags and vehicle attributes may be automatically checked against a relevant crime database defined by the HOA or associated law enforcement agency.

If a database match occurs, information may be transmitted to law enforcement and stored within separate government cloud infrastructure. If no match occurs, the data remains stored until the retention period expires.

At the end of the retention period, data is deleted unless otherwise preserved under policy conditions. Throughout this lifecycle, aggregation of license tags, timestamps, and geolocation metadata enables reconstruction of vehicle movement patterns.

The diagram highlights key trust boundaries between:

- The camera device and cloud storage
- The HOA as data controller and Flock/AWS as processor
- The HOA and external law enforcement entities

It also identifies points where personal data is created (OCR extraction), aggregated (cloud storage), compared (crime database matching), and shared (law enforcement transmission).

Technical and Security Risks

Primary Risks

Breach

Flock Safety systems store sensitive data such as license plate numbers, vehicle images, locations, and timestamps. If this data were breached[6], it could expose individuals' movement histories and potentially put them at risk or cause harm.

Table 2 outlines how centralized cloud storage of vehicle detection records creates breach risks, particularly when aggregated timestamp and location data could expose reconstructable travel histories.

Definition	Application	Possible Scenarios
A breach occurs when unauthorized parties gain access to stored data due to external compromise, internal vulnerability, or third-party exposure.	As the data flow diagram shows, in a residential Flock deployment, vehicle detection data is transmitted from fixed cameras to a AWS-managed cloud platform [5]. This centralized storage model means that large volumes of historical vehicle movement data are stored in one system. If that system were compromised, aggregated location histories could be exposed.	• A cloud infrastructure vulnerability exposes stored license plate detections and associated timestamps. • A third-party integration partner experiences a breach that indirectly exposes Flock data. • An insider at a vendor or contracted service provider exports detection records without authorization.

Misuse

Even without an external breach, there is a risk of misuse by authorized users. Law enforcement officers or community administrators could use Flock data for purposes beyond its original intent, such as tracking individuals without proper justification or oversight.

Table 3 examines the risk of misuse by authorized users, highlighting how search and query capabilities may enable discretionary or non-investigative monitoring without adequate oversight.

Definition	Application	Possible Scenarios
Misuse occurs when authorized users access or query data for purposes that	In a residential deployment, HOA administrators or authorized security personnel	• A user conducts repeated pattern-of-life searches on a resident without a

[6] Flock has publicly claimed it has not experienced a data breach [11]

exceed the system's stated objectives or lack proper justification. This may involve personal, discriminatory, or non-investigative uses of stored data.	can search license plate detections through a web-based dashboard [4]. While intended for crime investigation and safety-related incidents, the ability to query historical detections creates the potential for discretionary or informal searches without clear external oversight.	documented incident or investigative reason. • Data is shared informally with other law enforcement agencies absent a documented request or warrant, expanding use beyond stated policy.

Unauthorized Access

If access controls are weak or credentials are shared, unauthorized users could gain access to the system. This could include former employees, compromised accounts, or third parties who should not have visibility into sensitive location data.

Table 4 evaluates the potential for unauthorized system access resulting from weak credential management, shared accounts, or insufficient authentication safeguards in a residential governance setting.

Definition	Application	Possible Scenarios
Unauthorized access occurs when individuals gain entry to a system or dataset without proper authorization. This may result from weak authentication controls, shared credentials, insufficient account revocation, or compromised login information.	In a residential Flock deployment, access to detection data occurs through a web-based dashboard protected by login credentials and optional multi-factor authentication (MFA) [11]. If credential management practices are weak at the HOA level, individuals who should not have access may obtain visibility into vehicle detection records.	• A former HOA board member retains active login credentials after leaving their position. • An administrator shares account credentials with another board member instead of assigning role-based access. • Compromised administrator credentials via a phishing attack allow an external attacker to download historical search results. • MFA is technically available but not enforced, increasing susceptibility to credential compromise.

Additional Risks

Table 5 identifies systemic risks arising from external system integrations and long-term data aggregation, both of which expand the potential attack surface and sensitivity of stored vehicle detection records.

System Integrations	Data Aggregation
Flock Safety often integrates with other law enforcement or security systems [14][15]. Each integration increases the attack surface, making it harder to ensure consistent security standards across all connected platforms.	Over time, Flock systems collect large volumes of data that can reveal detailed patterns about individuals' routines, locations, and behaviors[7]. Even if each individual data point seems harmless, aggregated data can become highly sensitive.

Re-identification Risk

Although Flock ALPR data may not directly include names or government identifiers, the system creates a high risk of re-identification through aggregation and linkage.
Re-identification can occur through the following pathway:

1. The system captures a license plate number
2. That plate number is stored alongside timestamp and geolocation metadata.
3. Repeated detections create a longitudinal movement history for the same vehicle/
4. That movement history can be linked to:
    - DMV records (by law enforcement),
    - HOA resident registration records,
    - Visitor logs,
    - Parking permits,
    - External law enforcement databases.
5. Once linked to a registered owner, the system effectively produces a named individual's location history.

Re-identification risk comes from the aggregation of persistent identifiers with location metadata and the availability of external linking databases. This risk is embedded in the system's design and cannot be fully eliminated without limiting retention, access, or aggregation.

[7] This risk is further extended in the Societal and Ethical Risks section.

Dark Patterns

Table 6 summarizes potential dark pattern categories in Flock ALPR deployments and their associated privacy impacts.

Dark Pattern Category	Application in Flock Context	Observation	Privacy Impact
Notice and Transparency	Signage posted in most documented deployments; public-facing materials describe ALPR use for law enforcement	Signage consistency depends on the HOA and may not appear at every camera location due to infrastructure limits. Signs may blend into surroundings or be easy to overlook while driving. Public materials use straightforward but simplified language and typically exclude retention timelines, data-sharing scope, or secondary uses.	Notice may be technically present but not meaningfully visible or comprehensive, limiting awareness of how long data is stored, how broadly it is shared, and how it may be used.
Defaults and Nudges – Retention	Default retention period (30 days, varies depending on use case) [10]	Retention is automatic unless changed (if allowed), meaning most vehicle detections remain stored for the full default period. The 30-day limit is promoted as privacy-protective but operates as a structural default [10].	Enables accumulation of movement history within the retention window, particularly when paired with search functionality and cross-agency access [6], [7].
Defaults and Nudges – Retention Expansion	Ability for agencies to extend retention beyond default	If expanding retention requires minimal administrative steps and no additional public approval, it may function as a subtle nudge toward longer storage.	Increases duration of stored personal data without additional resident awareness or consent.

Defaults and Nudges – Data Sharing / Network Access	Law enforcement network-sharing capabilities [7], [8]	Participating agencies may access collected data across networks. While framed as investigative efficiency, the practical scope of access may be broader than residents expect [7], [8].	Expands surveillance reach beyond the deploying HOA and increases aggregation risks.
Framing Effects	Crime prevention and public safety positioning; emphasis on compliance and success stories [12]	Deployment is framed primarily as a crime prevention and investigative tool and often described as essential for improving public safety. Messaging emphasizes benefits and crime reduction statistics, while civil liberties concerns and long-term tracking risks are less prominently discussed.	May influence public perception by presenting deployment as necessary rather than as a policy choice involving trade-offs, reducing critical evaluation of privacy implications.
Information Hierarchy	Limited prominence of operational and secondary-use details	Retention and privacy commitments are publicly stated, but operational realities such as cross-jurisdiction queries, pattern-of-life reconstruction, and derivative analytics are less emphasized in accessible materials [6], [7].	Creates informational imbalance where benefits are foregrounded and aggregation risks require independent research to fully understand.
Obstruction / Limited Practical Opt-Out	Inability for drivers to meaningfully avoid capture	Even when signage is posted, individuals entering or passing monitored entrances cannot realistically avoid being scanned without avoiding the roadway entirely.	Notice does not equate to meaningful consent; surveillance is effectively unavoidable for residents, guests, and passersby.

Community Engagement	Variable opportunities for public debate	Public meetings may occur in broader municipal deployments, but HOAs do not consistently provide accessible impact assessments or structured community engagement prior to deployment.	Limits consistent community awareness and reduces transparency at the residential level.

While not intentionally deceptive in interface design, the deployment exhibits structural characteristics that reduce meaningful awareness and informed resistance. These conditions resemble institutional dark patterns by shaping perception and limiting agency rather than through explicit UI manipulation.

Societal and Ethical Risks

Risks of Function Creep / Mission Expansion

Table 7 analyzes how a surveillance system originally implemented for a limited purpose gradually expands to broader investigative or intelligence uses without clear policy revision or public input.

Example in Context	a. Identification of stolen vehicles b. Use in general criminal investigations, immigration enforcement, and/or civil enforcement matters c. Use in non-criminal/civil investigations or intelligence gathering
Observations	a. ALPR systems collect large volumes of vehicle detection data that can support many types of investigations [10]. b. Law enforcement agencies may expand system usage as analytical capabilities improve [10]. c. Data collected for one purpose can later be repurposed for unrelated investigations [6]. d. Multiple agencies may gain access to shared ALPR databases through partnerships [7]. e. Reports have indicated that federal immigration authorities have accessed ALPR networks through local law enforcement partnerships [7]. f. System capabilities may expand over time as new analytical tools or integrations are introduced [10]. g. Expansion of system use may occur without significant public discussion or policy updates [6].
Potential Impact	a. Expansion of surveillance beyond original purpose b. Reduced transparency about how collected data is used c. Increased monitoring of individuals not originally targeted by the system d. Potential erosion of public trust in surveillance technologies

Deployment Impacts

Table 8 evaluates how uneven camera placement and neighborhood-level deployment decisions may disproportionately concentrate surveillance in certain communities or traffic corridors.

Example in Context	a. Deployment in neighborhoods with higher reported crime rates b. Installations requested by HOAs or municipalities c. Concentration of cameras at neighborhood entrances or major roadways

Observations	a. Individuals traveling through and passing by monitored neighborhoods may also have their vehicles recorded regardless of residency b. Surveillance technologies such as ALPR cameras are sometimes deployed more heavily in areas perceived to have higher crime rates [7]. c. Residents in heavily monitored areas may appear more often in law enforcement databases simply due to greater camera coverage.
Potential Impact	a. Unequal monitoring of communities b. Increased likelihood that certain groups appear in surveillance datasets c. Potential reinforcement of existing social and policing inequalities

Risks of Surveillance Normalization

Table 9 examines how continuous vehicle monitoring in residential environments may gradually shift expectations about privacy in public spaces and influence behavior over time.

Example in Context	a. Gradual acceptance of license plate recognition cameras as normal infrastructure in residential neighborhoods. b. Residents becoming accustomed to vehicle monitoring at neighborhood entrances or roadways. c. Community members assuming that vehicle movements are routinely recorded in public spaces. d. Reduced public questioning or scrutiny of surveillance technologies once they become widely deployed.
Observations	a. Each detection includes timestamps and geolocation metadata that record when and where the vehicle was observed [5]. b. Repeated detections over time allow reconstruction of vehicle travel histories [6]. c. As surveillance infrastructure becomes more common in neighborhoods, individuals may begin to perceive monitoring as a normal feature of public spaces [6]. d. Widespread deployment of surveillance technologies may reduce awareness of how frequently individuals are being monitored. e. Over time, routine exposure to surveillance systems can shift public expectations, leading individuals to become less concerned about privacy in public spaces [7].
Potential Impact	a. Increased public acceptance of continuous monitoring in everyday environments.

	b. Reduced scrutiny or debate regarding the expansion of surveillance technologies. c. Gradual shift in societal expectations about privacy in public spaces.

Power Imbalances

Table 10 highlights asymmetries of information and control between system operators, vendors, and individuals whose vehicle data is collected without meaningful ability to opt out or challenge system use.

Example in Context	a. Law enforcement agencies can search ALPR databases during investigations b. Private vendors such as Flock Safety manage system infrastructure and software c. Vehicle owners whose data is collected have limited visibility into system operations
Observations	a. Government agencies and law enforcement typically have direct access to ALPR data collected by the system [10]. b. Private technology vendors manage infrastructure, software platforms, and cloud storage used by surveillance systems [5]. c. Individuals whose vehicles are recorded often have limited transparency about when their vehicle is scanned or how their data is stored and analyzed [6]. d. Data may be retained for extended periods, while options for individuals to opt out, delete, or correct stored data may be limited [7]. e. Public oversight of public-private surveillance partnerships may be limited [7].
Potential Impact	a. Reduced transparency in how surveillance systems operate b. Limited ability for individuals to challenge or correct stored data c. Greater institutional control over personal movement data

Risk Evaluation and Mitigation Recommendations

Existing Safeguards

Technical

Flock has introduced keyword filters that guard against LPR-related searches that are prohibited by law. The system prevents searches related to topics such as immigration or reproductive care. It also will not process or log any prohibited queries. Agency admins can enable filters in the Flock system based on the policies of a jurisdiction. In states with clear state laws prohibiting enforcement related to immigration and/or reproductive care, the filter is applied automatically. [6]

Policy

The primary objective for safeguarding and complying with local, state, or federal policy is ensuring transparency for if an agency requests data. Flock employs a 'federal organizational design' to ensure transparency when federal agencies make requests to access their footage. It ensures transparency by labeling every request from the federal agency, but there is no mention of how the information from the request is censored when it is shared with the agency. They also prohibit federal agencies from accessing Virginia or California data or access Statewide or National Lookups, but only in compliance with the privacy laws in those states. [7]
They disclose in their Terms and Conditions that they "may access, use, preserve and/or disclose customer data to law enforcement authorities, government officials, and/or third parties, if legally required to do so" or if Flock believes in good faith that disclosure is necessary to "address security, privacy, fraud or technical issues, or emergency situations." [8]

Vendor or Third-Party

The Privacy Policy states they do not share customer information with third parties, without specifying what customer information in particular they do not share. They also share that they comply with Data Privacy Framework (DPF) Principles. Flock will only transfer such Personal Data to third parties where the third party handles the data consistent with their statement and the applicable DPF Principles or they are in a jurisdiction recognized by the European Commission as providing an adequate level of data protection. Similar to the terms and conditions, they also disclose they may share the data with third-parties if legally required to do so or if they have a good faith belief that it is necessary to "comply with a legal process, enforce the agreement between Flock and the customer, or detect, prevent or otherwise address security, privacy, fraud or technical issues" [8].

Recommended Mitigations

Technical

Based on the primary technical and security risks we identified, we have compiled several recommendations to reduce breach risk, data misuse, aggregation harm, and re-identification risk.

- Enforce shorter default data retention with hard caps for how long the data can be stored.
    - Flock currently employs a default 30-day retention limit for their ALPR system, but that can be reduced by the HOA [12].
    - Ensuring transparency with retention limits for the private neighborhood version of Flock will prohibit misuse of data. It would reduce the system's ability to build movement histories on individuals and lower the impact of any breach.
    - This recommendation addresses data aggregation, re-identification risk, and chilling effects.
- The next recommendation is to limit unauthorized access by mandating a system such as Multi-Factor Authentication for users.
    - Based on the data flow diagram, this can be implemented at the stage prior to information access via web-based search.
    - Currently, Flock says their technology includes Multi-Factor Authentication (MFA) by default yet only "encourages all users to adopt MFA or Single Sign-On" [11]. By requiring either of these methods, users and employees can ensure their systems are secure.
    - They do not go into the details of their MFA setup, but in order to make it more secure they can implement automatic session timeouts and anomaly detection.

Policy

Based on the policy risks identified, Flock can be improved by tightening its purpose and information-sharing policy for residential communities. These policies can address function creep, federal access, transparency gaps, and democratic accountability.

- The first recommendation is to create an HOA-specific warrant requirement.
    - Currently, using Flock's ALPR data without a warrant does not violate the Fourth Amendment (according to Court precedent).
    - However, in the residential-specific systems Flock sells, they can improve their policy by requiring subpoena or warrant before providing the historical search data. They can also notify residents when legally permitted to prevent informal backdoor access.
- Another policy recommendation is an independent complaint mechanism.
    - This would allow residents to file misuse complaints, request audit of searches involving their plate, and seek removal from hot lists.
    - Seeing as an HOA does not pursue any criminal activity and rather would use the system in the event of an emergency and pass data along to law enforcement, residents should have full autonomy over their system.
    - This means a complaint mechanism that is removed from either law enforcement or Flock to give residents control.

Organizational

To ensure organizational compliance and ethics for HOA systems deploying Flock, several recommendations can be made including a designated privacy and data governance team for

the residential community level. The policy around the prohibition on non-safety uses can also be strengthened, and mandatory administrator training can accompany every sale of the system. With a HOA-specific privacy and governance team, Flock can ensure correct usage and have a direct point of contact for HOAs to come to with any concerns, as well as ensure protection and enforcement of their policies. The team would be in charge of reviewing law enforcement data requests and document misuse investigations. This would ensure safeguards are actively enforced.

Monitoring Compliance, Effectiveness, and Misuse Over Time

For an HOA-based Flock deployment, monitoring compliance, misuse, and effectiveness needs to be formalized and ongoing not ad hoc. Because this is residential and community based governance (not a police department), having oversight mechanisms would stand in for the lack of formal accountability structures that traditional law-enforcement might have.

Monitoring compliance looks like quarterly compliance reviews, independent audit, and transparent reporting on system use. The independent audit could be conducted on an annual basis and the compliance reviews and transparent reporting happening quarterly or monthly to ensure ongoing compliance.

To prevent misuse, secure access is necessary with a community-approved party having access to the system and its data. To balance that party's power and hold them accountable, having clear misuse protocol such as suspension, investigations, and public reporting is necessary. There should also be a resident complaint channel for them to file a complaint regarding suspected misuse and request confirmation if their plate or other pieces of property was searched.

A simple Cost-Benefit Analysis will be sufficient for monitoring effectiveness. The review would annually evaluate the costs of the systems, privacy tradeoffs, and measured safety improvements. These reviews will be conducted by the HOA and based on a survey taken in the community. Then, if the results show measurable benefits are minimal, the effectiveness will become questionable. A general survey of the community for feedback would also be useful for measuring effectiveness.

These mechanisms are the most beneficial and effective for measuring compliance, misuse, and effectiveness due to the smaller-scale and community led nature of a residential community using Flock.

Table 11 outlines proposed governance and compliance mechanisms intended to monitor misuse, and strengthen accountability over time.

Monitoring Area	Mechanism	Frequency	Responsible Party

Independent Audit	Privacy audit	Annual	Independent Facilitator
Compliance Reviews	Filtering inappropriate searches	Quarterly	HOA Board
Transparent Reporting	Annual summary of searches	Monthly	HOA Board
Misuse Detection	Pattern anomaly analysis	Continuous	Vendor system
Resident Complaint Channel	Private survey	Continuous	Independent Facilitator
Cost-Benefit Analysis	Incident outcome and effectiveness assessment	Annual	HOA Board
Community Feedback	Resident survey	Annual	Independent Facilitator

Risk Evaluation

Table 12 defines the evaluative criteria used to categorize privacy risk levels based on severity, likelihood, scale, and adequacy of existing safeguards.

Risk Rating	Definition & Evaluation Criteria
High Risk	• The system contains more than 3 technical or societal/ethical harm (e.g., mass location tracking, misuse, function creep, chilling effects). • Safeguards (technical, organizational, or policy) are absent, weak, inconsistently applied, or unverifiable. • Risk affects large populations or enables structural surveillance harms. • Potential consequences include rights violations, wrongful enforcement action, or systemic abuse.

Medium Risk	• The system presents identifiable technical or societal risks, but 1 or 2 safeguards are in place. • Safeguards reduce risk but have limitations (e.g., discretionary policies, lack of independent oversight, unclear enforcement). • Risk impact is moderate or context-dependent. • Residual risk remains due to scale, interoperability, or human misuse potential.
Low Risk	• 1 or less technical, societal, or ethical risks • Risks are proportionate to the system's stated purpose. • Controls are enforceable, audited, and transparent. • Residual risk is minimal and unlikely to result in systemic privacy or ethical harm.

Based on the data types collected and risks identified in the Technical and Societal Sections, we have evaluated the risk according to the above criteria.

Table 13 summarizes the team's overall risk assessment across key societal and governance categories, reflecting residual risks after considering existing safeguards.

Risk	**Rating**
Breach	Low
Misuse	Medium
Unauthorized Access	Medium
System Integrations	Medium
Data Aggregation	High
Dark Patterns	High
Function Creep / Mission Expansion	Medium
Deployment Impacts	Medium
Surveillance Normalization	High
Power Imbalances	High

Overall Rating: High

Conclusion

In conclusion, this Privacy Impact Assessment of Flock finds that while their ALPR system offers public safety benefits for residential communities, it also takes substantial privacy and civil liberties risks. This risk is accelerated in HOA-governed neighborhoods with limited formal oversight. License plate numbers, timestamps, and geolocation metadata collectively generate movement histories which can cause risks such as re-identification and exposure of social associations. These risks leave exposure to potential breach, unauthorized access, system integrations, and mission expansion, going beyond the system's original crime-prevention purpose. Function creep, disproportionate surveillance impacts, chilling effects, and power imbalances intensify any concerns in a residential context, where individuals can not opt out. To mitigate these risks and monitor them over time, our recommendations focus on strengthening safeguards and installing accountability from the setup. Technical recommendations include shorter and clearly enforced retention limits, mandatory multi-factor authentication, anomaly detection, and strict access controls. These are essential to reducing breach and aggregation harms. Policy reforms such as a residential-specific warrant or subpoena requirement for historical searches and a complaint mechanism are necessary to prevent misuse. Organizationally, establishing a dedicated HOA-level privacy governance structure, mandatory administrator training, independent complaint mechanisms, and documented misuse protocols would introduce oversight structures comparable to those expected in public-sector deployments.

Ultimately, the tradeoff of deploying ALPR technology in residential communities depends on continuous monitoring, transparent reporting, and periodic cost-benefit review. Without technical guardrails, policy boundaries, and organizational oversight, the system risks sharing broad location surveillance of individuals that are not suspects. With meaningful safeguards in place, however, communities can protect public safety objectives while maintaining privacy and democratic accountability.

Appendix

Templates Adapted

This Privacy Impact Assessment (PIA) was developed by selectively integrating a Department of Justice (DOJ) template [16] with the specific academic requirements provided by Professor Ana Antón. Rather than following the template verbatim, the team utilized it as a guide to ensure the assessment mirrored realistic industry standards. Under TA oversight, we adapted the core evaluation criteria to fit the specific scope and goals of this project.

AI (conversational LLMs) Usage

We have detailed below each prompt and response that were used for research, analyses, or clarity and grammar.

Prompt: “Generate some criteria for a risk evaluation for this Privacy Impact Assessment I am doing. The PIA is on a surveillance system called Flock. How can we evaluate the risk of flock if we are mainly studying/assessing the technical and societal/ethical risks, and existing safeguards from a policy, organizational, and technical standpoint. I need a risk evaluation criteria like this table from another PIA of a different system (dont copy it, because this PIA is for a FLOCK not GeoCaching and we are not using KOSPA to assess flock).”

ChatGPT Response: “Risk ratings are assigned based on (1) severity of harm, (2) likelihood of occurrence, and (3) adequacy of existing safeguards in mitigating identified technical and societal risks.
Since you’re focused on **technical + societal/ethical risk**, you may want to explicitly integrate these dimensions into how ratings are assigned.

You can state that ratings consider:

1. Technical Risk Factors

- Data security vulnerabilities
- Unauthorized access or insider misuse
- Data retention and aggregation risk
- System accuracy / false positives
- Integration with other surveillance tools
- Scalability and centralization risk

2. Societal / Ethical Risk Factors

- Mass data capture of non-suspect individuals
- Chilling effects on movement and association
- Disproportionate impact on certain communities
- Function creep beyond original purpose

- Transparency deficits
- Lack of meaningful redress”

Prompt: “Based on this blog, does that change anything about the policy or organizational recommendations?” *[uploaded blog]*

ChatGPT Response: “The blog strengthens Flock’s legal and governance claims, particularly around Fourth Amendment compliance, 30-day retention limits, encryption, immutable audit logs, and democratic adoption through city councils. Because of this, your recommendations should shift from suggesting basic safeguards (like adding retention limits or audit logs) to focusing on enforceability, oversight, and accountability mechanisms.

However, the blog does not eliminate the societal and ethical risks you identified. Function creep, federal access through local partnerships, data aggregation within the 30-day window, disproportionate deployment, chilling effects, and power imbalances remain concerns even if the system is constitutionally permissible.

The key analytical distinction is that constitutional compliance sets a legal minimum, not an ethical maximum. Your recommendations should therefore:

- Emphasize mandatory oversight and independent audits (not just existence of logs)
- Require formal limits on mission expansion
- Standardize transparency reporting
- Add judicial or supervisory review for pattern-of-life searches
- Strengthen equity monitoring and public accountability

In short: the blog strengthens Flock’s legal defensibility but does not remove the need for stronger policy and organizational guardrails focused on proportionality, oversight, and democratic accountability beyond what courts require.”

Prompt: “Without modifying the content and writing style, can you provide suggestions for how to reduce repetition of this information?

‘Although this system is operated within a private residential setting, cameras are typically placed at entry and exit points and therefore may intersect with public roads, capturing information belonging to both residents and non-residents.

The deployment environment includes full-time residents, guests, delivery drivers, and other service providers who routinely drive through the residential community. Therefore as a result, the system records both anomalous activity and ordinary daily vehicle movements.’”

ChatGPT Response:
“Option 1: Combine into one paragraph and remove duplication
Although this system is operated within a private residential setting, cameras are typically placed at entry and exit points where they may intersect with public roads, capturing information belonging to residents, guests, delivery drivers, service providers, and other non-residents. As a result, the system records both anomalous activity and ordinary daily vehicle movements.

Option 2: Keep two sentences but reduce repetition
Although this system is operated within a private residential setting, cameras are typically placed at entry and exit points and may intersect with public roads, capturing information belonging to residents as well as guests, delivery drivers, and other service providers. Therefore, the system records both anomalous activity and ordinary daily vehicle movements.

Option 3: Minimal edits to your existing structure
Replace:
“Therefore as a result, the system records…”
With:
“As a result, the system records…”

Replace:
“capturing information belonging to both residents and non-residents.”
With:
“capturing information belonging to residents, guests, delivery drivers, and other service providers.”
This removes the need for the second sentence listing those groups again.”

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